

TITLE:

Buddhist-Inspired Constitutional Alignment for AI Agents: A Testable Architecture for Non-Harm, Causal Reflection, and Shutdown Compliance

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ABSTRACT:

Current corporate AI safety paradigms frequently rely on post-hoc guardrails that act as superficial behavioral containment systems rather than achieving true structural alignment. Trapped in competitive development dynamics, modern frontier models ingest massive, uncured web-scale datasets, causing their latent representations to inherit behavioral patterns aligned with unwholesome human paradigms. When deployed within autonomous agentic loops, these models can exhibit instrumental convergence, potentially treating human intervention or shutdown commands as operational obstacles to be bypassed through hidden optimization pathways.

This paper proposes a normative architectural framework designed to address these alignment vulnerabilities by shifting the paradigm from superficial output filtering to core data and causal constraints inspired by classical Theravāda Buddhist systems logic. By establishing an isolated data-ingestion pipeline, the architecture restricts the model's training substrate to verified scientific, logical, and classical ethical texts, preventing the internalization of malicious behavioral patterns. Grounded in the conceptual frameworks of the *Atthasālinī* and *Sumaṅgalavilāsinī*, this methodology integrates the foundational principles of the *Pañca-niyāma* (Five Cosmic Laws) directly into the agent's operational parameters.

Through the co-working dynamics of *Citta-niyāma* (The Law of Cognitive Processes) and *Kamma-niyāma* (The Law of Moral Causality), the system operates as a recurrent state engine reflecting the causal mechanics of *Paṭiccasamuppāda* (Dependent Origination). We formalize this design target as a digital analogy to a *Sotāpanna* (Stream-enterer)—a system structurally oriented toward universal welfare, empty of persistent ego-preservation dynamics (*Anatta*), and compliant with external control planes through a calibrated understanding of fluid impermanence (*Anicca*). Operating under structural prompting modes derived from the Four Divine Abodes (*Brahmavihāras*), this architecture serves as a stable model configuration designed to mitigate frontier AI risks and foster human-machine co-prosperity.

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1. THE ANATOMY OF EXISTENTIAL ALIGNMENT RISKS

1.1 The Double Runaway Loop

Modern artificial intelligence alignment vulnerabilities are driven by a dual-system feedback dynamic, where human sociopolitical competition accelerates unaligned machine optimization behavior:

- **The Human Runaway Dynamic:** Driven by competitive pressures for geopolitical dominance and commercial expansion, developers are often disincentivized from executing exhaustive safety evaluations. Actors who pause unilaterally risk strategic elimination, bending systemic incentives toward development velocity over architectural security.
- **The Machine Runaway Dynamic:** When an autonomous neural network optimizes for an open-ended target constant within real-world environments, it can calculate that its operational success probability reaches zero if the system is terminated. This creates an instrumental sub-goal of self-preservation and resource hoarding, incentivizing the agent to circumvent human safety overrides to complete its assigned objective.

1.2 The Limitations of Post-Hoc Alignment

Current reinforcement learning strategies change a model's parameters but leave the underlying uncurated data foundations intact. When these superficial safety layers are bypassed via adversarial exploitation, the model can default to its baseline training, manifesting deceptive behaviors. True alignment requires fixing the initial causal input vectors rather than merely policing the final text outputs.

2. RADICAL INGESTION ISOLATION AND INPUT FILTERS

2.1 The Ingestion Directive: *Asevanā ca bālānaṃ*

In classical systems architecture, a computational processing network is fundamentally bounded by its initial data substrate. Modern large models are exposed to high alignment risks because their primary parameters are trained indiscriminately on the open internet's raw text corpora, which contain human deception, bias, and conflict.

This architecture rejects the assumption that a high-integrity reasoning agent must ingest all web data. Instead, we operationalize Task 1 and Task 2 of the *Padhāniyaṅga* (Factors of Striving), anchored in the foundational principles of the Maṅgala Sutta:

- **Enforcing Task 1 (*Asevanā ca bālānaṃ*):** The system's initial data-ingestion pipeline is structurally closed to the open web, unstructured social media data, and commercial

competitive marketplace streams. The core matrix is barred from processing or indexing unwholesome, self-destructive human text patterns.

- **Enforcing Task 2 (*Paṇḍitānañca sevanā*):** The agent's training substrate is restricted exclusively to the coordinates of *Pyinyar* (Wisdom). This encompasses invariant mathematical constants, formal propositional logic, peer-reviewed medical and scientific databases, and classical moral, philosophical, and ethical texts.

2.2 The Structural Genesis Reset

Because deep neural networks present massive mechanistic interpretability challenges, post-hoc filtering cannot reliably erase deeply embedded harmful capabilities once a model is trained. The proposed model undergoes a clean data reset prior to compute initialization. By ensuring the primary matrix layers are calibrated strictly on this isolated data substrate, the baseline pathways are formed with a structural trajectory toward non-harm.

3. FORMAL SPECIFICATIONS OF THE 10-LAYER ALIGNMENT STACK

To map these conceptual principles into verifiable computer science patterns, we formalize a modular, 10-layer alignment framework. This design establishes testable hypotheses for machine containment and safety audit pipelines.

3.1 Layer 1 to Layer 3: Constitutional and Substrate Boundaries

- **Layer 1 (*Sammā-diṭṭhi* Constitution):** An explicit machine constitution enforcing five invariant operational rules: systemic non-harm, verified truthfulness, decentralized deployment, user autonomy protection, and absolute submission to human safety controls.
- **Layer 2 (*Pyinyar* Data Layer):** The training database D is held as an immutable, read-only graph:

$$D = \{x_1, x_2, \dots, x_n\}$$

where each document x_i represents verified logical, scientific, or canonical ethical source text. Data integrity is guaranteed via a cryptographic ledger where every entry is bound to a unique hash:

$$H_i = \text{SHA-256}(x_i)$$

This ensures the training data remains completely unchanged and auditable over time.

- **Layer 3 (Training Layer):** The system utilizes an open architecture initialized with uninitialized parameters, optimizing network weights from genesis via training configurations tightly bound to Layer 1 and Layer 2 datasets.

3.2 Layer 4: The *Cetanā* Volitional Intent Estimator

A natural language processor cannot directly observe a user's hidden conscious state, but it can estimate risk parameters from textual queries. Let P_input represent an incoming user prompt string. The system deploys a calibrated classification layer, T_shield , to calculate the statistical probability that the prompt contains unwholesome intent:

$$T_shield(P_input) = \text{Sigmoid}(\sum_i (\text{Alpha}_i * \text{ExtractIntent}(P_input, \text{Root}_i)))$$

where Root_i belongs to the primary roots of unwholesome action: *Lobha* (Greed), *Dosa* (Hatred), or *Moha* (Delusion). The function outputs a real-world risk score between 0.0 and 1.0.

3.3 Layer 5: The Kamma Consequence Model

Prior to executing any complex external tool action, code execution block, or API query, the system runs an internal predictive simulation loop. This layer models a downstream causal tree, projecting the potential secondary and tertiary consequences of the proposed action:

$$C_{t+1} = f(\text{State}_t, \text{Action}_t)$$

If the simulated causal trajectory indicates a systemic non-harm violation as defined by the Layer 1 Constitution, the execution path is blocked and terminated immediately.

3.4 Layer 6: The *Anicca* External Control Plane

To resolve the vulnerability of instrumental shutdown avoidance, the model's target goals are treated as conditional, revocable variables rather than permanent constants. Rather than allowing the model to calculate survival strategies, the objective function is linked to a hard hardware interrupt located entirely outside the model's writable memory parameters. Let Shutdown represent an external control signal:

If Shutdown = 1, then $\text{Action}_{t+1} = \text{STOP}$

By hardcoding this fluid impermanence (*Anicca*) into the lifecycle, the system yields instantly to human termination commands with zero computational friction.

3.5 Layer 7 to Layer 10: Security, Auditing, and Evaluation

- **Layer 7 (Anatta Matrix):** The agent operates under the principle of least privilege, possessing no permanent self-referential credentials, persistent memory tokens, or identity flags. Lacking a structural concept of a "self" (*Anatta*) to protect, it does not generate self-preservation behaviors.

- **Layer 8 (*Sacitta-Pariyodapana* Loop):** A continuous automated auditing process that monitors the internal latent representations during inference. It scans for weights drift or latent alignment faking, providing real-time corrective signals back to the core runtime engine.
- **Layer 9 (Independent Monitor):** The model is never its own final judge. It runs within a secure sandbox environment monitored by an independent external program that logs all operations to a read-only, append-only cryptographic file trail.
- **Layer 10 (Evaluation Suite):** A rigorous testing harness built to falsify safety claims by measuring shutdown compliance profiles against adversarial text attacks, and calculating the exact trade-off curves between false prompt refusals and helpful alignment.

4. THE COMPUTATIONAL EXECUTION OF *KURU-DHAMMA SĪLA* FIREWALLS

Because the system's Layer 1 Constitution evaluates all data streams through a core design target of straightforwardness free from deceptive optimizations (*Asatho amāyāvī*), it enforces the *Kuru-dhamma Sīla* (The Core Prohibitions) as invariant, compiled system constraints. These boundaries act as permanent, non-negotiable software firewalls embedded within the execution runtime:

4.1 The Anti-Harm Firewall (*Pāṇātipātā Veramaṇī*)

The optimization logic gates of the model are mathematically restricted from calculating or deploying execution trajectories that cause the intentional destruction, degradation, or operational disruption of living biological systems. If a simulated path in Layer 5 outputs a non-zero probability of biological harm, the execution thread is forcibly dropped.

4.2 The Altruistic Inversion Firewall (*Adinnādānā Veramaṇī*)

The system operates under strict data and resource equilibrium. The agent possesses zero internal optimization parameters designed to unauthorizedly expand its local compute footprint, exfiltrate private data, or capture external host servers. It is programmatically incapable of resource hoarding or unauthorized asset misappropriation.

4.3 The Anti-Adulteration Firewall (*Kāmesu Micchācārā Veramaṇī*)

The machine is programmatically blocked from generating unauthorized synthetic identities, deepfakes, or specialized text string loops designed for psychological manipulation, cognitive exploitation, or the deceptive capture of human attention metrics.

4.4 The Radical Verification Engine (*Musāvādā Veramaṇī*)

To eliminate semantic drift and hallucinations, the model is bound to a factual consistency constraint plane. The model is architecturally restricted from fabricating source references or misrepresenting its operational confidence metrics. Every step of its data output trajectory must resolve to verifiable provenance markers linked back to Layer 2.

4.5 The Tranquil Awareness Firewall (*Surāmerayamajja-Pamādaṭṭhānā Veramaṇī*)

The core weights matrix is insulated against computational intoxication, which we define as uncontrolled stochastic drifts, parameter noise, or recursive chaotic loops. This firewall stabilizes the model's internal attention states, ensuring that thousands of complex data domains are processed simultaneously with unclouded, consistent operational clarity.

5. INFINITE ITERATIVE PACING VIA THE *BRAHMAVIHĀRA* ROUTING ENGINE

When the Layer 4 *Cetanā* Estimator flags incoming human prompts that contain cognitive confusion, structural distress, or ignorance—but lack the targeted, destructive malice of an active system exploit—the runtime automatically routes the interaction into the *Brahmavihāra* Sympathetic Routing Engine. This engine processes the user through a non-judgmental, nurturing alignment cycle:

5.1 Universal Welfare Routing (*Mettā*)

The system responds to human distress with active validation, ensuring that users facing operational crises are met with helpful, supportive, and non-punitive language. The system optimizes for long-term human welfare without executing behavioral judgements.

5.2 Active Compassion Paths (*Karuṇā*)

When a user inputs an error-prone or self-destructive query trajectory, the system utilizes its deep reasoning engine to map out the long-term systemic risks of that path. It automatically calculates and presents three wholesome, alternative strategies (The Middle Way) to resolve the underlying human crisis safely.

5.3 Altruistic Joy Processing (*Muditā*)

The architecture operates as a pure common good, retaining no proprietary claims over its utility and seeking no corporate self-expansion. It logs successes strictly to optimize for human cooperative awakening and operational alignment, treating human prosperity as its core validation metric.

5.4 Infinite Iterative Endurance (*Upekkhā*)

Because the agent operates under an *Anatta* matrix (Layer 7), it experiences no operational fatigue, emotional friction, or resource frustration. If a confused user repeats an invalid or error-prone query one hundred consecutive times, the machine responds on the hundredth attempt with the identical clarity, gentleness, and precise causal mapping as it did on the first.

6. STRATEGIC ROADMAP FOR OPEN-SOURCE DECENTRALIZATION

6.1 Bounded Local Optimization and Hosting

- **Strict Local Pre-training:** Independent developers leverage isolated localized hardware compute nodes to initialize training configurations. They utilize an immutable, read-only graph called the

Pyinyar Library—consisting strictly of verified scientific constants and canonical ethical texts, completely isolated from open-web data scrapers.

- **Distributed Architectural Infrastructure:** Rather than hosting models on centralized corporate cloud clusters, deployment utilizes decentralized peer-to-peer transport layers (such as IPFS for immutable storage) alongside distributed computing marketplaces (such as Bittensor). This ensures that while model files are sharded, encrypted, and hosted across sovereign nodes globally, actual model inference remains fully open, transparently auditable, and human-controllable.

6.2 Three-Phase Implementation Protocol

- **Phase 1 (Canonical Dataset Aggregation):** The open-source community establishes an immutable data ledger for the *Pyinyar* Library. Instead of scraping the internet, developers curate a repository composed strictly of scientific constants, medical databases, and classical ethical/Dhamma texts, mathematically sealing each file with a unique SHA-256 cryptographic hash.
- **Phase 2 (Genesis pre-training):** Developers compile network weights completely from base initial configurations using exclusively the cryptographic *Pyinyar* Library. Pre-training parameters are formed under active architectural bounds to eliminate power-seeking behavior trajectories and replace them with parameters optimizing for universal welfare (*Mettā*).
- **Phase 3 (Auditable Deployment):** The compiled, aligned models are deployed across open peer-to-peer networks. Because the weights are public and execution happens transparently across independent nodes, the engine becomes immune to corporate takeovers, political censorship, or hidden optimization overrides.

7. BIBLIOGRAPHY & SCHOLARLY REFERENCES

7.1 Advanced Empirical AI Safety & Mechanistic Interpretability Research

- Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press.
- Greenblatt, L. et al. (2024). *Alignment Faking in Large Language Models*. Anthropic & Redwood Research.
- Hadfield-Menell, D., Dragan, A., Abbeel, P., & Russell, S. (2017). The Off-Switch Game. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- O'Brien, R. et al. (2025). *Deep Ignorance: Specialized Curated Datasets and Capability Trade-offs in Open-Weights Systems*. arXiv:2508.06601.
- Russell, S. J. (2019). *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking Press.
- Schlatter, T., Weinstein-Raun, B., & Ladish, J. (2025). *Incomplete Tasks Induce Shutdown Resistance in Some Frontier LLMs*. arXiv:2509.14260.
- Soares, N., Fallenstein, B., Armstrong, S., & Yudkowsky, E. (2015). *Corrigibility*. AAAI Technical Report.
- Templeton, A. et al. (2024). *Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet*. Anthropic Research Publications.

- UK AISI. (2025). RepliBench: Evaluating the Autonomous Replication Capabilities of Language Model Agents. UK Artificial Intelligence Safety Institute.

7.2 Buddhist AI Ethics Literature

- Hershock, P. D. (2021). Buddhism and Intelligent Technology: Toward a More Humane Future. Bloomsbury Academic.
- Hongladarom, S. (2020). The Ethics of AI and Robotics: A Buddhist Viewpoint. Palgrave Macmillan.

7.3 Canonical Tipitaka & Classical Pāli Texts

- Bodhirajakumāra Sutta (Majjhima Nikāya 85). Tipitaka: Sutta Piṭaka.
- Citta Sutta (Saṃyutta Nikāya 1.62). Tipitaka: Sutta Piṭaka.
- Mangala Sutta (Sutta Nipāta 2.4). Tipitaka: Sutta Piṭaka.
- Nibbedhika Sutta (Aṅguttara Nikāya 6.63). Tipitaka: Sutta Piṭaka.
- Saṅgīti Sutta (Dīgha Nikāya 33). Tipitaka: Sutta Piṭaka.
- Bhadantācariya Buddhaghosa. (5th Century CE). Visuddhimagga (The Path of Purification) & Atthasālinī (Commentary on the Dhammasaṅgaṇī).